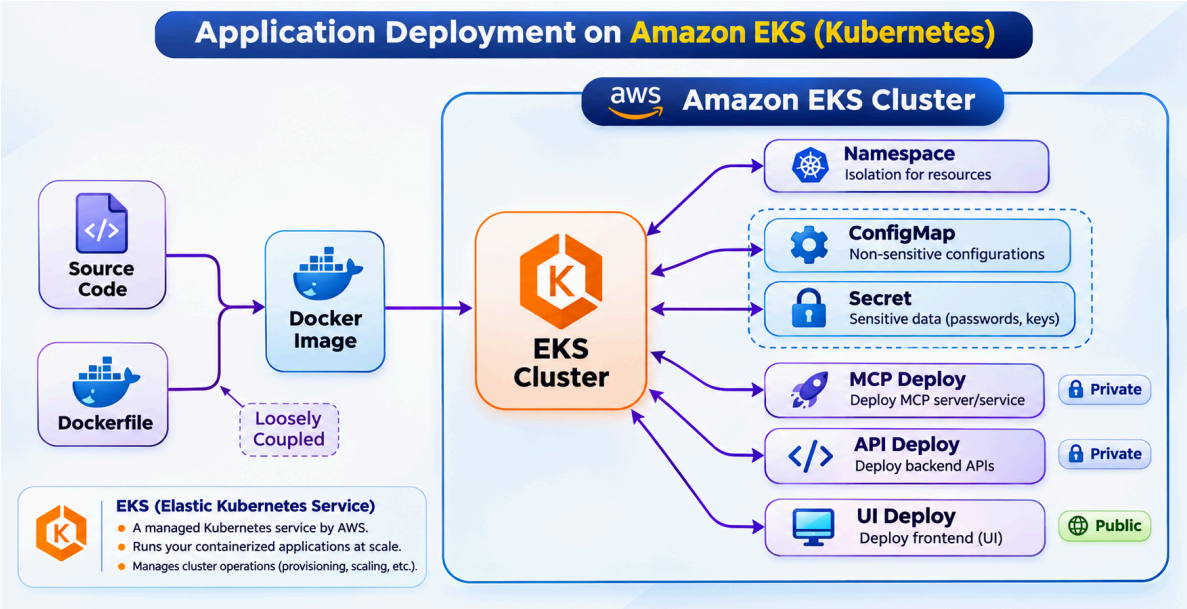
Banking Agentic AI application deployment with Docker & K8S



Step-1 : Create Dockerfile for the application (Ex: Dockerfile)

```
FROM python:3.12-slim

WORKDIR /app

# 1. Install libraries (cached unless requirements.txt changes)
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# 2. Copy project code (.dockerignore skips .env, venv, k8s ...)
COPY . .

# 3. Default command = FastAPI. K8s overrides it for the MCP server and the UI.
CMD ["uvicorn", "app.api.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Step-2 : create docker image and push into docker registry (docker hub)

```
$ docker build -t ashokit/bankbuddy:1.0

$ docker images

$ docker login

$ docker push ashokit/bankbuddy:1.0
```

Step-3 : Create EKS Cluster

Steps : https://github.com/ashokitschool/DevOps-Documents/blob/main/05-EKS-Setup.md

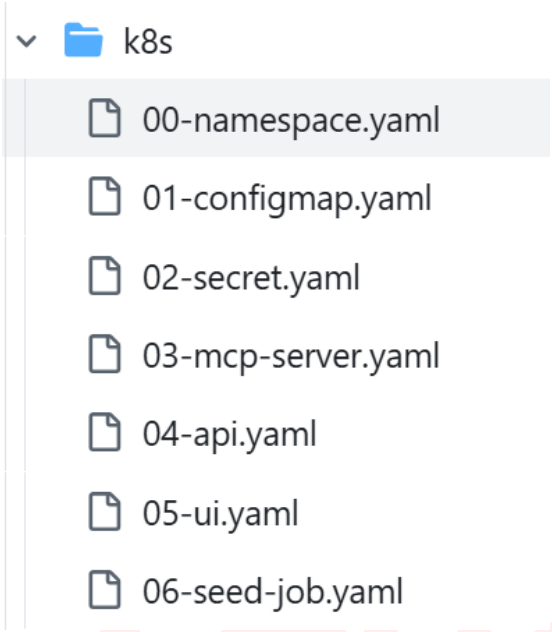
Step-4 : Create AWS RDS Instance and note down DB Properties

DB Endpoint URL : database-1.XXXXXX.ap-south-1.rds.amazonaws.com
DB Username : root
DB Password: root4321
DB Port : 3306
DB Name : bank_db

Note: Enable 3306 port in RDS Instance Security Group Inbound Rules.

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Step-5 : Create K8S manifest ymls and configure required details(db props, openai api key, docker image name in respective file given below)



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Step-6 : Execute k8s ymls using kubectl command

```
$ kubectl apply -f .

$ kubectl get all -n bankbuddy

$ kubectl get pods -n bankbuddy
```

Step-7 : Connect with RDS instance using mysql workbench and verify data loaded or not.

Step-8 : Enable Node Port Number (30080) in Worker Nodes Security Group inbound rules.

Step-9 : Access streamlit ui application using worker node public ip.

http://node-public-ip:30080/